



Maven Toys Sales & Inventory Analysis Project

Data-driven insights to optimize toy sales, streamline inventory, and boost store performance.

by - Sonali Aruk

Project Summary

This project analyzes sales, inventory, product, and store data for a multi-location retail chain specializing in toys, games, and art & crafts. The goal is to solve key business challenges such as stockouts, overstocking, profitability tracking, and store performance monitoring.

Using four key datasets 4 Sales, Products, Inventory, and Stores 4 the project delivers data-driven insights and actionable recommendations.



Project Objectives

Inventory Optimization:

Identify low-stock and overstock situations to minimize missed sales opportunities and reduce excess inventory costs.

Sales Performance Analysis:

Track top-selling products, categories, and regional trends to optimize marketing and stocking strategies

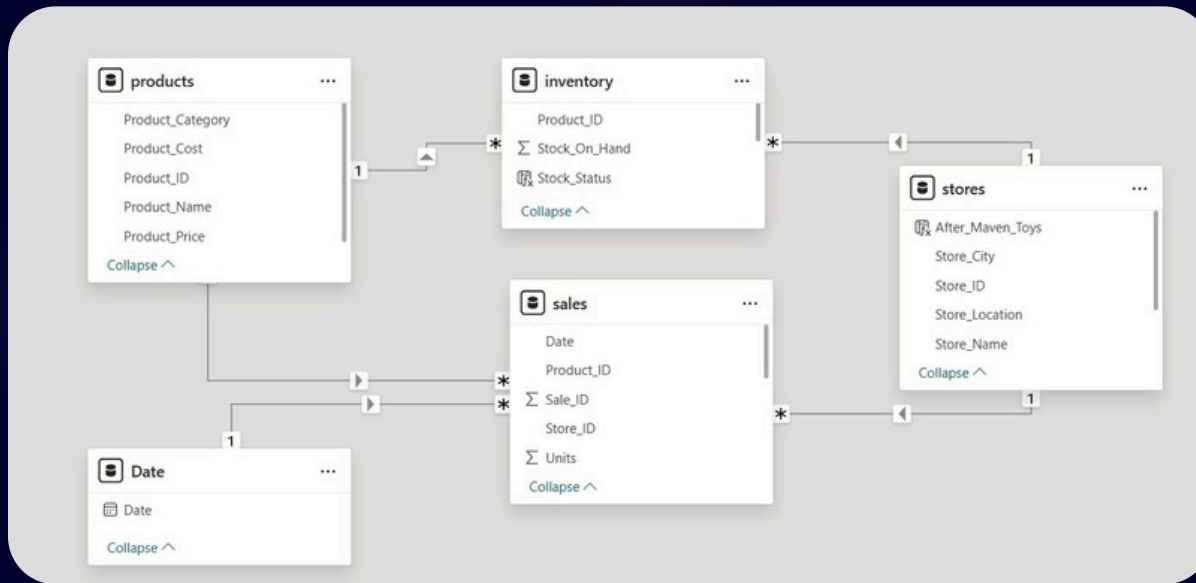
Profitability Insights:

Calculate product-level and category-level profit margins to focus on high-value items.

Store Performance Monitoring:

Compare stores by revenue, profit, and operational metrics to identify under performing locations and growth opportunities.

Data Architecture Overview



This model has **two fact tables (Sales and Inventory)** sharing common dimensions like Products and Stores.

It represents a **fact constellation schema**, allowing separate analysis of sales performance and stock levels.

Core Tables:

- **Products** - details, categories, pricing
- **Sales** - unit sold, dates, store_id, product_id
- **Inventory** - Stock levels, store_id, product_id
- **Stores** - Location , store_name, open_date..
- **Date**

Primary keys and foreign key relationships ensure data integrity while enabling complex multi-dimensional analysis

Relationships :

- Products & sales - one to many
- product & inventory - one to many
- stores and sales - one to many
- stores and inventory - one to many
- Date and Sales - one to many

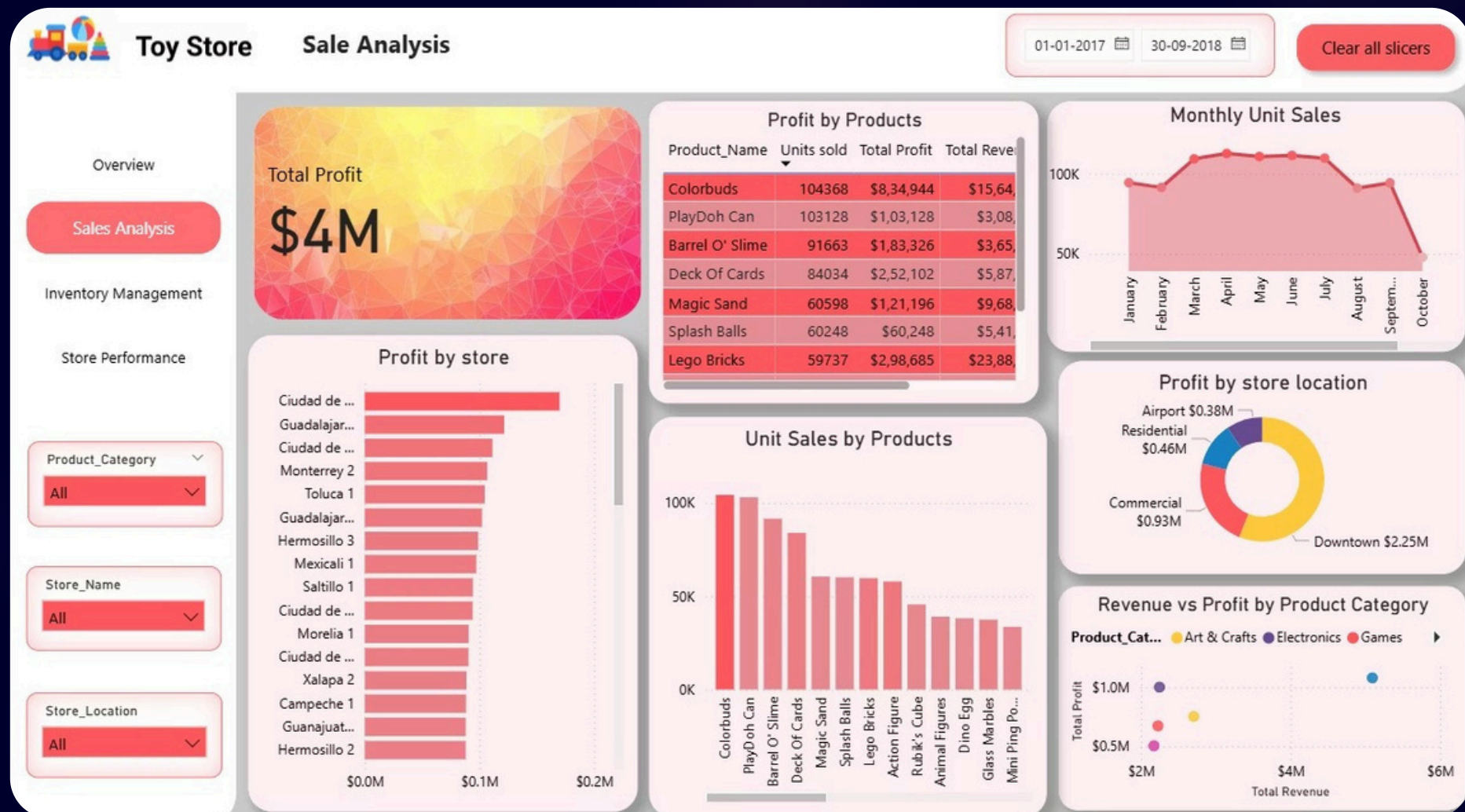
Page 1 - Overview

- \$14.44M revenue, \$4.01M Profit, 27.79% profit margin.
- Toys & Electronics are top categories.
- Colorbuds is the leading product.
- Steady revenue growth, but 5% stockouts exist.
- average daily revenue has increased by 30% compared to the same period last year, showing positive year-over-year growth.



Page 2 - Sales Performance Analysis

- \$4M profit.
- Top products: Colorbuds, Barrel O9 Slime which sell more.
- Top stores: Ciudad de México, Guadalajara.
- Downtown , Commercial stores contribute most profit.
- Electronics category revenue not much but profit maximum.
- jenga & playfoam give more profit margin.



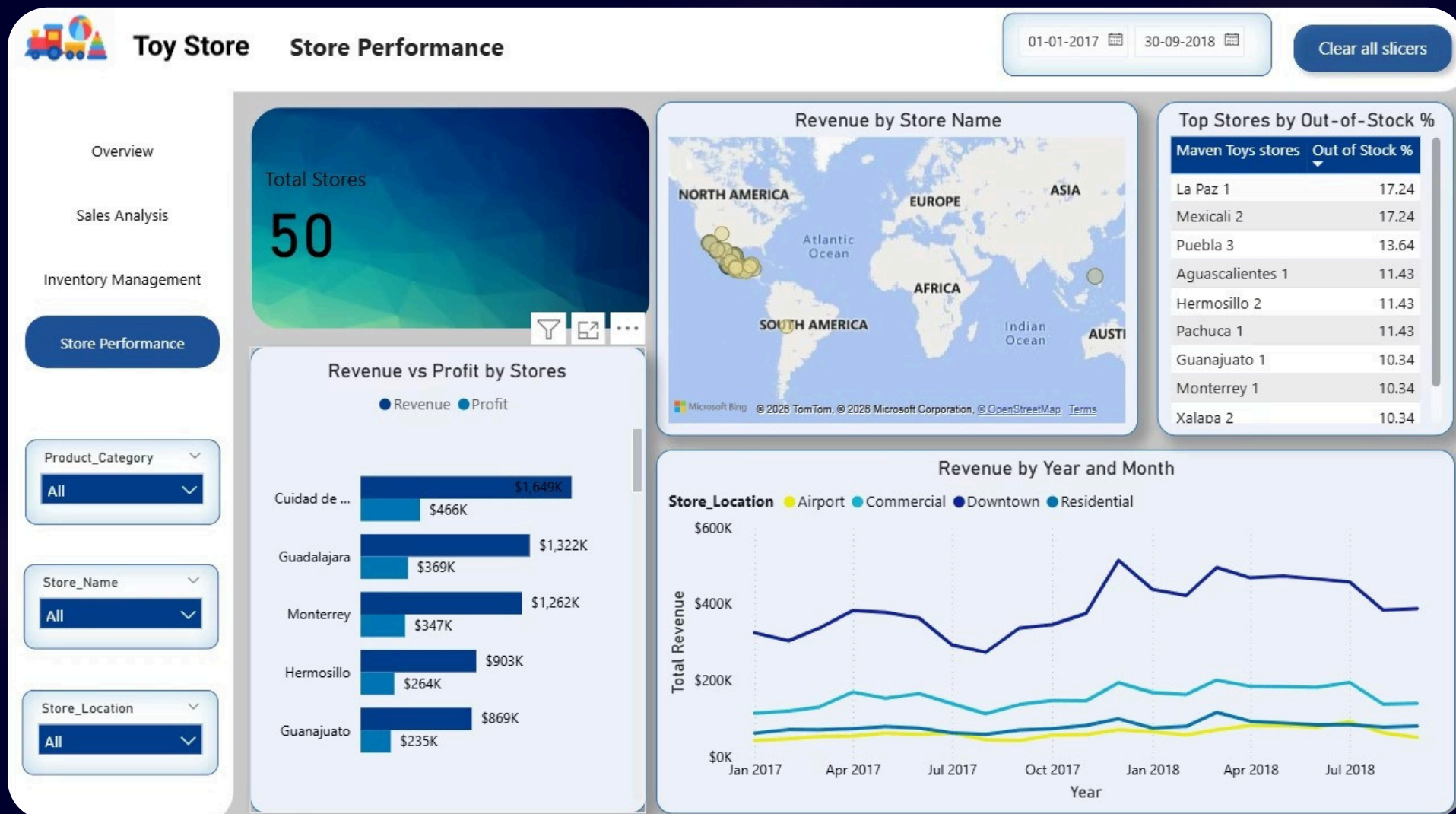
Page 3 - Inventory Management

- 30K units instock, stockouts - 5%
- Deck of Cards, Dinosaur Figures, PlayDoh 4 highest stock.
- Turnover improved (2017 - 2018).
- Art & Crafts and Toys hold most stock focus on electronics.



Page 4 - Store Performance

- 50 stores total, strong in Mexico's major cities.
- Top stores : Ciudad de México, Guadalajara, Monterrey.
- Weak: Saltillo, Morelia, Santiago.
- La Paz 1 & Mexicali 2 stores have high stockouts.
- Downtown stores dominate in revenue.



Strategic Recommendations

Inventory Automation

An AI system that monitors stock levels and **alerts us in advance** when items are running low, so we can take timely action and avoid shortages

Premium Product Focus

Sell more high-profit products that customers like the most .

Marketing

Run **seasonal promotions** to boost sales. Start a **loyalty program** to keep customers coming back

Leverage top-performing categories and seasonal sales trends to design **localized campaigns**, boosting revenue by **10-15%**.

Store Operations

Give more support to stores that perform well.
Change products in each store based on local customer needs.

Thank you !